



HOW life

WHERE AND HOW LIFE BEGAN are questions that nobody really knows the answers to. Scientists have a number of theories but Earth has undergone so many changes that there is no longer any firm evidence for exactly how it started. Life could have stopped and started several times before conditions became stable enough for living things to survive.

Toxic planet



Earth formed about 4.6 billion years ago. In the beginning it was a hot, molten rock surrounded by poisonous gases and deadly radiation. Gradually, it cooled and a surface crust formed. Volcanoes spat out gases from the core, filling the atmosphere with carbon dioxide, nitrogen, and water vapor. As the cooling continued, the water vapor turned into liquid and rained down to form the oceans. Although Earth was still a hostile place, it was ready for life to start.

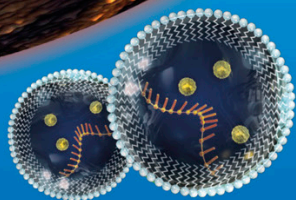


CHEMICAL SOUP

Life probably began in the oceans. All the elements in living things—carbon, hydrogen, nitrogen, oxygen, phosphorus, and sulphur—were present in the atmosphere, but in a different mixture from today. Lightning bolts caused chemical reactions. These formed simple chemicals that washed into the oceans, where they reacted with others to form more complex molecules. Some of these molecules developed an ability to copy themselves. Once they could do that, the development of life speeded up.

GROWING A THICK SKIN

These self-copying molecules were fragile and needed protection from the harsh conditions. One type of molecule, a phospholipid, could form into bubbles, trapping copier molecules inside them. The bubble acted as a barrier, allowing the molecules inside to produce new substances more easily. These were the first cells—the basic units of life.



EARLY CELLS

